

Yu SHUAI

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RESEARCH INTERESTS

Fluid dynamics; model reduction; data-driven machine learning; uncertainty quantification; flow control.

EDUCATION

Princeton University | Princeton, NJ, USA

Aug 2022–Present

Ph.D. candidate in Mechanical and Aerospace Engineering

Advisor: Clarence W. Rowley | Expected defense: July 2027

Peking University | Beijing, China

Sep 2018–Jul 2022

B.Sc. in Theoretical and Applied Mechanics

PUBLICATIONS

Published

1. Y. Shuai, C. Liu, and D. F. Gayme. “Structured input–output analysis of oblique laminar–turbulent patterns in plane Couette–Poiseuille flow.” *International Journal of Heat and Fluid Flow*, 103, 109207 (2023). doi: [10.1016/j.ijheatfluidflow.2023.109207](https://doi.org/10.1016/j.ijheatfluidflow.2023.109207)
2. C. Liu, **Y. Shuai**, A. Rath, and D. F. Gayme. “A structured input-output approach to characterizing optimal perturbations in wall-bounded shear flows.” *2023 American Control Conference (ACC)*, pp. 2319–2325 (2023). doi: [10.23919/ACC55779.2023.10156058](https://doi.org/10.23919/ACC55779.2023.10156058)
3. Y. Shuai, C. Liu, and D. F. Gayme. “Structured input-output analysis of oblique turbulent bands in transitional plane Couette–Poiseuille flow.” *Proceedings of the 12th International Symposium on Turbulence and Shear Flow Phenomena (TSFP12)*, Osaka, Japan (2022). [Proceedings PDF](#)

Submitted

4. Y. Shuai and C. W. Rowley. “Symmetry-reduced model reduction of shift-equivariant systems via operator inference.” *Advances in Computational Mathematics* (submitted; in revision). [arXiv:2507.18780](https://arxiv.org/abs/2507.18780) (2025).
5. Y. Shuai and C. W. Rowley. “CONS²D: Coupled optimization for non-intrusive learning of subspaces and symmetry-reduced dynamics for shift-equivariant systems.” In preparation.

RESEARCH EXPERIENCE

Doctoral Researcher | Princeton University

Aug 2022–Present

Mechanical and Aerospace Engineering, Princeton, NJ, USA | Advisor: Clarence W. Rowley

- **Symmetry-reduced operator inference.** Developed reduced models that learn the shape dynamics of spatially shifting solutions in a co-moving frame and reconstruct the shift amount; validated on the Kuramoto–Sivashinsky equation.
- **Coupled subspace–dynamics learning.** Developed CONS²D — Coupled Optimization for Non-intrusive learning of Subspaces and Symmetry-reduced Dynamics — which learns the projection subspaces and the symmetry-reduced dynamics together by trajectory optimization, data-driven and non-intrusive, without knowledge of the full-order operators. On the non-reciprocal Cahn–Hilliard equation it improves on the SR-Galerkin and SR-Oplnf benchmarks and generalizes more reliably than other current trajectory-optimized models; the learned subspaces explore more directions than the benchmarks do. On a three-dimensional linearized Navier–Stokes example it matches intrusive balanced POD at equal subspace dimension on unseen test data.
- **Lagrangian coherent structures from sparse tracer data** (with Nicholas Conlin and Marcus Hultmark). Proposed a pipeline for detecting Lagrangian coherent structures in the atmospheric surface layer from sparse tracer trajectories: a large-eddy-simulation baseline, 4D-Var estimation and calibration against the measured trajectories, then geometric LCS detection. A baseline velocity-field reconstruction by nearest-neighbour interpolation is in place; the large-eddy simulation and the 4D-Var step are in progress.

Undergraduate Researcher | Peking University

Oct 2021–Jun 2022

School of Mathematical Sciences, Beijing, China | Advisor: Zhifei Zhang

Undergraduate thesis. Studied transient growth and transition thresholds for Couette–Poiseuille flow using asymptotic analysis and Chebyshev-collocation eigenvalue computations.

Visiting Student Researcher | Johns Hopkins University

May 2021–Oct 2021

Mechanical Engineering, Baltimore, MD, USA | Advisor: Dennice F. Gayme

Applied structured input–output analysis to transitional Couette–Poiseuille flow, characterizing oblique turbulent bands, dominant wavelengths and Reynolds-number scaling.

Undergraduate Project Lead | Peking University

Sep 2020–Nov 2021

State Key Laboratory for Turbulence and Complex Systems, Beijing, China | Advisor: Yipeng Shi

Developed wavelet-based, self-similar reconstruction of corrupted rotating-turbulence data and compared it with gappy POD and GAN baselines.

RESEARCH SOFTWARE

SROplnf — Python package for symmetry-reduced operator inference and POD–Galerkin models, with reproducible Kuramoto–Sivashinsky examples and tests.**SELECTED PRESENTATIONS**

79th APS Division of Fluid Dynamics Meeting | Orlando, FL, USA

Nov 22–24, 2026

Scheduled oral presentation: “Inferring Lagrangian Coherent Structures in the Atmospheric Surface Layer from Sparse Particle Measurements.”**SIAM Conference on Mathematics of Data Science (MDS26) | Salt Lake City, UT, USA**

Nov 16–20, 2026

Scheduled. Minisymposium co-organizer (with Clarence W. Rowley) and oral presentation: “Non-Intrusive Optimization of Reduced-Order Models for Shift-Equivariant Systems based on Symmetry Reduction” in the minisymposium *Data-Driven Nonlinear Reduced-Order Models for Transport-Dominated Systems*.**78th APS Division of Fluid Dynamics Meeting | Houston, TX, USA**

Nov 23–25, 2025

Oral presentation: “Non-Intrusive Model Reduction of Shift-Equivariant Systems via Data-Driven Projection and Symmetry-Reduced Dynamics.”

SIAM Conference on Applications of Dynamical Systems | Denver, CO, USA

May 11–15, 2025

Minisymposium co-organizer (with Clarence W. Rowley) and oral presentation: “Reduced-Order Modeling of Shift-Equivariant Systems Using Operator Inference” in the minisymposium *Reduced-Order Modeling of Fluid Flow with Coherent Structures*.**77th APS Division of Fluid Dynamics Meeting | Salt Lake City, UT, USA**

Nov 24–26, 2024

Oral presentation: “Neural Network-Based Closure Model of the Ensemble-Averaging Dynamics of Turbulent Puffs in Transitional Pipe Flow.”

75th APS Division of Fluid Dynamics Meeting | Indianapolis, IN, USA

Nov 20–22, 2022

Oral presentation: “Structured Input-Output Analysis of Oblique Laminar-Turbulent Flow Patterns in Transitional Plane Couette-Poiseuille Flow.”

TEACHING

Assistant in Instruction, Princeton University:

- **MAE 305: Mathematics in Engineering I** Fall 2026
- **MAE 222: Mechanics of Fluids** Spring 2026
- **MAE 305: Mathematics in Engineering I** Fall 2025
- **MAE 553: Turbulent Flow** Spring 2025
- **MAE 433: Automatic Control Systems** Fall 2024

HONORS AND AWARDS

Guggenheim Second Year Fellowship, Princeton University	2023
Award for Excellent Undergraduate Research Program, Peking University	2021
Huatai Financial Technology Scholarship, Peking University	2021
Boeing Scholarship, Peking University	2020

TECHNICAL SKILLS

Programming Python (NumPy, SciPy, JAX, PyTorch), MATLAB, Fortran, C++**CFD libraries** [Dedalus](#), [Channelflow 2.0](#), [Openpipeflow](#)**High-performance computing** MPI, Slurm